CircleHub – Technical Presentation Guide

Slide-by-Slide Comprehensive Reference & Architecture Manual for Panel Defense

Live Production URL: https://frontend-seven-black-12.vercel.app

---

■ Slide 1: Project Title & Executive Overview

Project Title: CircleHub – Hyper-Local Community Networking & Resource Sharing Platform

Domain: Full-Stack Web Development / Cloud-Native Social Networking Infrastructure

Core Objective: Connect neighbors, empower local interest groups, coordinate community events/games, enable secure multi-format file/photo sharing (up to 25MB), and facilitate WhatsApp-based social invitations with strict phone privacy protection.

■ Slide 2: Tech Stack Architecture & Complete API Reference

Layer	Technologies & Tools Used
Frontend UI/UX	React 18, Vite 5, HTML5 Canvas API, Custom CSS3 Design System (Glassmorphism, Flexbox/Grid)
Backend API	Node.js, Express.js (REST API Router & Serverless Handler), Body-Parser (50MB Limit)
Database	MongoDB Atlas (Cloud NoSQL Document DB), Mongoose ODM with Indexed Schemas
Deployment Cloud	Vercel Edge Cloud Network (SPA Hosting + Serverless API Functions)

■ Third-Party External APIs

• **WhatsApp Universal Web & Share API (https://api.whatsapp.com/send):** Formats and transmits event details, circle invitations, and game photo links into WhatsApp chat messages with direct website URLs.

• **Google Maps Search & Directions API (https://www.google.com/maps/search/):** Encodes full event venue addresses into dynamic Google Maps navigation links.

■ Internal CircleHub RESTful API Endpoints (/api)

Category	Endpoint Method & Path	Description & Functionality
Auth	POST /api/register POST /api/login	User registration and authentication
Profile	GET /api/users/:id PUT /api/users/:id	Fetch and update profile, mobile number, privacy preference, and avatar photo
Circles	GET /api/circles POST /api/circles POST /api/circles/:id/join POST /api/circles/:id/leave GET /api/circles/:id/members DELETE /api/circles/:id	Circle creation, discovery, membership operations, and privacy-filtered member lists
Resources	GET /api/circles/:id/resources POST /api/circles/:id/resources DELETE /api/circles/:id/resources/:resId	Retrieve, upload (up to 25MB), and delete shared files & photos for a circle
Events	GET /api/events POST /api/events POST /api/events/:id/rsvp POST /api/events/:id/photos DELETE /api/events/:id	Event hosting, RSVP tracking, and post-game gallery photo uploads
Analytics	GET /api/dashboard/:userId	Dynamic user statistics and recommended local circles
Health	GET /ping, GET /	Server keep-alive and MongoDB status checks

■ Slide 3: Core Algorithms & Engineering Methods

1. **Client-Side Canvas Image Optimization & Compression Algorithm:** Utilizes HTML5 <canvas> to inspect image dimensions, downscale high-resolution photos (from 15MB+ camera photos to max 1920x1080), and compress to WebP/JPEG. Reduces payload size by up to 90% while maintaining visual clarity.
2. **Dynamic Aggregation Pipeline (MongoDB Analytics):** Uses MongoDB $match, $group, and $in pipelines to calculate live dashboard counts per logged-in user without hardcoded values.
3. **Multi-Level Privacy Enforcement Protocol:** Applies a granular access matrix (Everyone, Circle Members Only, Nobody) across backend serialization and frontend components for mobile numbers.
4. **Resilient Multi-Tier API Fallback Connection Method:** Client API layer utilizes AbortController (60s timeout for cold-starts) and automatic fallback URL redirection to guarantee sub-second availability.
5. **Direct Browser File Streaming Method:** Leverages HTML5 Blob streaming (URL.createObjectURL) to download shared circle resources (PDFs, Images, Docs) directly from memory.

■ Slide 4: Functional Modules & Key Features

• **Interest Circles & Resource Hub:** Community group creation and discovery across categories. Features a full Resource Sharing Hub where members upload and download files/photos up to 25MB.

• **Community Events & Game Photos:** Event creation with Google Maps navigation links, RSVP tracking, and post-game photo gallery with expanded Lightbox view.

• **WhatsApp Sharing & Direct Invites:** One-click WhatsApp sharing that dynamically embeds direct website URLs (https://frontend-seven-black-12.vercel.app) into invite messages.

• **User Profile & Phone Privacy:** Profile photo management (choose from device with auto-compression or remove photo) and customizable phone number privacy controls.

■ Slide 5: System Architecture & Data Flow

The system follows a modern cloud-native decoupled SPA architecture:

[React 18 SPA Frontend]
Vite 5 Build • Vercel Edge CDN Hosting

■■ *HTTPS / REST API Requests (JSON)* ■■

[Express.js API Server / Vercel Serverless Functions]
Body-Parser 50MB Limit • Route Authorization • Canvas Compression

■■ *Mongoose ODM Connection Pool* ■■

[MongoDB Atlas Cloud Database]
Indexed Collections: User, Circle, CircleMember, Event, Resources, Photos

■ Slide 6: Database Schemas & Data Model

• **User Collection:** name, email, password, location, bio, interests, skills, phone, phone_privacy ('Everyone'|'Circle Members'|'Nobody'), avatar.

• **Circle Collection:** name, description, category, location, created_by (ref User), resources[] (subdocuments: title, url, file_type, file_name, uploaded_by_name, created_at).

• **CircleMember Collection:** circle_id (ref Circle), user_id (ref User), joined_at.

• **Event Collection:** circle_id (ref Circle), title, description, event_date, location, created_by, attendees[], photos[].

■ Slide 7: Unique Selling Propositions (USP) & Innovations

1. **100% Dynamic Synchronization:** No mock or hardcoded statistics; counts update immediately across all active screens.
2. **Client-Side Compression:** Eliminates server payload crashes (413 errors) while supporting photos and files up to 25MB.
3. **Granular Privacy Controls:** User-driven contact visibility ensures privacy and safety in public communities.
4. **Seamless WhatsApp Social Loop:** Direct interactive website links embedded into WhatsApp shares drive viral user engagement.

■ Slide 8: Future Roadmap & Presentation Defense Tips

Future Enhancements: WebSockets (Socket.io) for live circle group chat, QR Code verification for event check-ins, and Web Push notifications for meetup reminders.

Panel Q&A; Defense Strategy:

• *Q: How do you prevent large image uploads from crashing the server?*
A: We implement client-side canvas compression to shrink images before transmission and set the Express body limit to 50MB.

• *Q: How is user privacy protected when sharing phone numbers?*
A: Mobile numbers are filtered on the server and client according to the user's privacy configuration (Everyone, Circle Members, Nobody).

• *Q: How does the application scale?*
A: Frontend and serverless functions are deployed on Vercel's global Edge CDN, paired with MongoDB Atlas cloud database.